# SOC Network Analysis & Incident Report

**Incident Title:** Network Connectivity and Port Enumeration – Google Connectivity Test

**Incident Type:** Network Diagnostics / Port and Service Enumeration

**Severity:** Informational – Low

**Status:** Initial Analysis Completed

**Analyst Role:** Tier 1 SOC Analyst

**Environment:** Kali Linux Virtual Machine

**Network Interface:** `eth0`

## 2. Executive Summary

A network connectivity and reconnaissance assessment was performed from a Kali Linux virtual machine to evaluate connectivity, routing, and exposed network services.

The system successfully completed a connectivity request and returned an IP address associated with Google's network. The reported average response time was 47.161 ms. A port scan identified TCP port 53, associated with the DNS service, and returned the service fingerprint `tcpwrapped`. Another scan reported no open ports or detected services.

No packet traffic or observed IP addresses were recorded during the packet-capture exercise. Consequently, there is insufficient packet-level evidence to determine whether suspicious network activity occurred.

**Initial assessment: No confirmed security incident was identified from the supplied results. Further validation is recommended to clarify the DNS service finding and verify the packet-capture configuration.**

## 3. Network Configuration and Connectivity

| Parameter | Observed Result |
|---|---|
| Local IP address | `10.0.2.15` |
| Network interface | `eth0` |
| Default gateway | `10.0.2.2` |
| Connectivity request | Successful |
| Average response time | 47.161 ms |
| Returned IP address | `172.217.27.78` |
| Reported hop count | 30 |

### Analysis

- **Local IP and gateway:** The addresses `10.0.2.15` and `10.0.2.2` are consistent with a typical VirtualBox NAT network configuration, although the hypervisor configuration has not been independently verified.
- **Connectivity:** The successful request indicates that the tested destination was reachable using the command or method employed.
- **Response time:** The average latency of 47.161 ms is an observation, not by itself an indication of a network performance issue or attack.
- **Returned IP:** `172.217.27.78` is within address space commonly used by Google. The exact destination and DNS resolution should be confirmed from the original command output.
- **Hop count:** The reported value of 30 requires validation. If this was a traceroute result, it may represent the maximum hop limit rather than the actual number of hops to the destination.

## 4. Port and Service Enumeration

### Scan Finding A

| Parameter | Result |
|---|---|
| Open ports | None reported |
| Detected services | None reported |

**Assessment:** No open ports or services were reported by this scan. This does not establish that the host has no listening services, because scan coverage, target selection, firewall filtering, and scan options affect the result.

### Scan Finding B

| Parameter | Result |
|---|---|
| Port | TCP/53 |
| Service | `domain` (DNS) |
| Service fingerprint | `tcpwrapped` |

**Assessment:** TCP port 53 is associated with the Domain Name System (DNS). The `tcpwrapped` label generally indicates that the scanner could not fully identify the service because the connection was closed, restricted, or otherwise limited. It does not independently confirm malicious activity or prove that a DNS server is operating normally.

The difference between the two scan results may be due to different target IPs, scan parameters, protocols, or network filtering. The original Nmap commands and target addresses are required to reconcile these findings.

## 5. Packet Capture and Traffic Analysis

| Parameter | Result |
|---|---|
| Interface selected | `eth0` |
| Traffic observed | None reported |
| IP addresses observed | None reported |
| Additional service information | TCP port 53; service identified as `tcpwrapped` |

### Analysis

No packet traffic or IP addresses were recorded in the supplied packet-capture results. This may indicate that no matching traffic was generated during capture, that the capture filter excluded relevant packets, or that the capture was started on an interface that did not carry the expected traffic.

The absence of captured packets is not evidence that the network was free of attacks. Additional capture validation is required before drawing a security conclusion.

## 6. Security Assessment

**Initial classification: Informational – No confirmed incident.**

### Observations

1. The tested connectivity request succeeded.
2. TCP port 53 was reported in one scan, with the service fingerprint `tcpwrapped`.
3. A separate scan reported no open ports or services.
4. The packet-capture exercise produced no reported traffic or IP addresses.
5. No direct indicators of compromise, confirmed unauthorized access, or malicious payloads were provided.

### Potential Concerns

- The TCP/53 finding should be checked against the intended target and expected network configuration.
- The discrepancy between scan results should be resolved before determining the target's exposure.
- Packet-capture visibility must be confirmed before relying on the results for threat detection.

These items are validation tasks, not confirmed security incidents.

## 7. Recommended Actions

### Immediate validation

1. **Confirm the target IP address.** Review the original commands to ensure that connectivity testing, Nmap scanning, and packet capture refer to the intended host.
2. **Validate the Nmap result.** Run a targeted scan against the authorized lab target and record the exact command, protocol, port state, and service-detection output.
3. **Check packet capture.** Confirm that `eth0` is active and that the capture is running while known test traffic is generated.
4. **Validate DNS behavior.** If the target is expected to provide DNS, perform an authorized DNS query and compare the result with the TCP/53 scan finding.
5. **Preserve evidence.** Save the complete command output, timestamps, scan options, and packet capture file for review.

### Suggested validation commands

```
ip addr show eth0
ip route
ping -c 4 172.217.27.78
traceroute -m 30 172.217.27.78
sudo nmap -sV -p 53 <authorized-target-IP>
sudo tcpdump -ni eth0
```

Replace `<authorized-target-IP>` with the intended lab target. Only scan systems for which you have permission. The Google connectivity test is separate from the lab-target port scan; do not assume Google is the target of the TCP/53 finding.

To validate packet capture, run `sudo tcpdump -ni eth0` in one terminal and generate authorized test traffic in another. Stop the capture with `Ctrl+C` after sufficient packets have been collected.

## 8. Analyst Knowledge Assessment

**Why do some commands require `sudo`?**

Some diagnostic commands require elevated privileges to access raw network sockets, capture packets, or inspect protected system resources.

**What is the importance of Nmap in cybersecurity?**

Nmap helps analysts identify reachable hosts, open ports, exposed services, and potential attack surfaces. Its findings support asset discovery, vulnerability assessment, and security validation.

**How can tcpdump help monitor attacks?**

Tcpdump captures and filters network packets, allowing analysts to investigate connection attempts, unusual traffic patterns, suspicious source addresses, and potentially malicious communications. It can reveal network activity that is not obvious from application-level information, but it does not automatically identify every attack or hidden service.

## 9. Conclusion

The supplied results demonstrate basic network connectivity and an initial attempt at port enumeration and packet capture. TCP port 53 was reported with a `tcpwrapped` fingerprint, while another scan found no open ports. The packet-capture exercise did not provide usable traffic evidence.

No security incident can be confirmed from the current evidence. The next step is to validate the scan targets and options, reproduce the port scan, and capture known test traffic before assigning a definitive security disposition.

**Recommended ITSM disposition: Close as Informational only if the lab exercise is complete and the results are explained. Otherwise, retain the ticket as Pending Validation until the port discrepancy and packet-capture visibility have been resolved.**